

Acemoglu & Restrepo (2018)

The race between automation and new tasks

Alvaro Marcelo Chávez Unyen
Repository 6 ù September 24, 2026

github.com/amchavezu/ai-06-restrepo NBER Working Paper 22252, revised June 2017. Proposition numbering follows this

version.

The aggregate economy becomes a task allocation problem

PAPER Final output combines a unit measure of tasks:

$$Y = \tilde{B} \left(\int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}.$$

- ▶ I is the automation frontier.
- ▶ N is the frontier of newly created tasks.
- ▶ $n = N - I$ measures the task range left to labor.

Automation

Capital can perform more existing tasks.

Reinstatement

New tasks expand the set in which labor has comparative advantage.

INTERPRETATION The unit of analysis is the aggregate task frontier, not an individual worker.

Source: Acemoglu and Restrepo, NBER revision, Sections 1 and 2.1

Effective costs determine who performs each task

PAPER All tasks can use labor. Tasks $i \leq I$ can also use capital.

$$p(i) = \begin{cases} \min\{R, W/\gamma(i)\}^{1-\eta}, & i \leq I, \\ (W/\gamma(i))^{1-\eta}, & i > I. \end{cases}$$

Cost threshold

Equilibrium threshold

$$\frac{W}{R} = \gamma(\tilde{I})$$

$$I^* = \min\{I, \tilde{I}\}$$

INTERPRETATION Assumption 1 makes $\gamma(i)$ strictly increasing. Capital takes the low-index tasks and labor takes the high-index tasks.

Source: Equations (5) and (6), NBER pp. 7–8

The task threshold becomes an endogenous production share

PAPER Under Assumptions 1, 2, and 3, Proposition 1 gives a unique static equilibrium:

$$Y = \frac{B}{1-\eta} \left[(I^* - N + 1)^{1/\hat{\sigma}} K^{\frac{\hat{\sigma}-1}{\hat{\sigma}}} + \left(\int_{I^*}^N \gamma(i)^{\hat{\sigma}-1} di \right)^{1/\hat{\sigma}} L^{\frac{\hat{\sigma}-1}{\hat{\sigma}}} \right]^{\frac{\hat{\sigma}}{\hat{\sigma}-1}}.$$

- ▶ Raising I^* increases capital's task share.
- ▶ Raising N creates a task at the top and removes one from the bottom.
- ▶ The factor shares change because technology reallocates tasks.

Source: Proposition 1 and equation (12), NBER pp. 9–10

Automation and new tasks push labor in opposite directions

PAPER In the technology-constrained region $I^* = I < \tilde{I}$:

	More automation $dI > 0$	More new tasks $dN > 0$
Relative wage W/R	Falls	Rises
Labor share	Falls	Rises
Employment	Falls	Rises
Productivity	Rises	Rises
Wage level W	Ambiguous	Rises

INTERPRETATION Productivity growth does not identify who receives the gains. The task margin separates the wage from output and the labor share.

Source: Propositions 2 and 3, NBER pp. 11–13

The wage sign is a race between two exact terms

Set $dN = 0$ in Proposition 3:

$$\frac{d \ln W}{dl} = \underbrace{\frac{B^{\hat{\sigma}-1}}{1-\hat{\sigma}} \left[\left(\frac{W}{\gamma(I^*)} \right)^{1-\hat{\sigma}} - R^{1-\hat{\sigma}} \right]}_{P_I \text{ productivity}} - \underbrace{(1-s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L}}_{D_I \text{ displacement}}.$$

Automation raises the wage if and only if $P_I > D_I$.

DERIVATION P_I measures the cost saving at the marginal automated task. D_I measures the pressure from concentrating labor in fewer tasks.

INTERPRETATION When the effective wage $W/\gamma(I^*)$ is close to R , the productivity gain is small and displacement dominates.

Source: Proposition 3 and discussion, NBER pp. 12–14

A numerical check makes the threshold visible

DERIVATION Normalize $R = B = 1$ and use illustrative values $\hat{\sigma} = 1.5$, $s_L = 0.6$, $\varepsilon_L = 0.5$, and $\Lambda_I = 2$.

Effective cost ratio c	P_I	D_I	$d \ln W / dI$
1.10	0.093	0.400	-0.307
1.5625	0.400	0.400	0.000
2.00	0.586	0.400	0.186
4.00	1.000	0.400	0.600

INTERPRETATION The sign changes at $c = (W/\gamma(I^*))/R = 1.5625$. This table evaluates the decomposition. It is not a calibration of the economy.

Source: Own computation in `analysis/wage_decomposition.py`

Capital adjustment changes the long-run wage result

PAPER Along an interior BGP, the Euler equation pins down

$$R = \rho + \delta + \theta g.$$

For $n > \bar{n}(\rho)$, a permanent increase in automation lowers $n = N - l$.

Outcome	Short run	Long run
Wage	May fall or rise	Rises
Employment	Falls	Falls
Labor share	Falls	Falls
Rental rate	Rises	Returns to $\rho + \delta + \theta g$

INTERPRETATION The wage can recover because capital accumulates. That recovery does not restore employment or labor's income share.

Source: Proposition 5 and Figure 6 discussion, NBER pp. 18–20

Scientists choose the direction of technology

Scientists allocate effort to automation or new-task creation:

$$\dot{I} = \kappa_I S_I, \quad \dot{N} = \kappa_N S_N, \quad S_I + S_N \leq S.$$

The task gap moves according to relative innovation values. An interior BGP requires

$$\kappa_I v_I(n) = \kappa_N v_N(n).$$

Automation value

Cost saving from replacing labor with capital at the bottom of labor's task range.

New-task value

Cost saving from replacing the capital-intensive alternative with labor at the frontier.

Source: Equations (22), (28), and (29), NBER pp. 21–25

Proposition 6 maps four balanced-growth regimes

PAPER Assume $\gamma(i) = e^{Ai}$, Assumption 2, $\hat{\sigma} > \zeta$, and $S < \bar{S}$.

Parameter region	BGP task gap	Outcome
$\rho < \bar{\rho}$	$n = 0$	Full automation
$\rho > \bar{\rho}, \kappa_I/\kappa_N > \bar{\kappa}$	$n \in (\bar{n}(\rho), 1)$	Unique interior BGP
$\rho > \bar{\rho}, \underline{\kappa} < \kappa_I/\kappa_N < \bar{\kappa}$	Interior	Multiple BGPs
$\rho > \bar{\rho}, \kappa_I/\kappa_N < \underline{\kappa}$	$n = 1$	No automation

INTERPRETATION The “race” has an interior solution only for part of the parameter space. Labor redundancy remains a genuine equilibrium possibility.

Source: Proposition 6, NBER pp. 25–27

The interior race is stable because prices redirect research

At the unique interior BGP,

$$\kappa_I v_I(n) = \kappa_N v_N(n).$$

- ▶ A fall in n means automation has run ahead.
- ▶ The resulting factor-price response lowers the return to further automation.
- ▶ New-task research becomes relatively more attractive.
- ▶ The task gap returns to its interior BGP when the unique-equilibrium condition holds.

PAPER With $\theta = 0$, the interior BGP is globally saddle-path stable. With $\theta > 0$, it is locally unique and asymptotically saddle-path stable.

INTERPRETATION A temporary automation shock can unwind. A permanent increase in κ_I/κ_N instead moves the economy to a new BGP with lower employment and labor share.

My analytical and computational work

Analytical check

- ▶ Isolated the exact wage condition $P_I > D_I$.
- ▶ Checked the signs rather than repeating the headline that automation hurts labor.
- ▶ Separated the short-run fixed-capital result from the BGP response.

Computational check

- ▶ Evaluated Proposition 3 over a grid of marginal cost ratios.
- ▶ Recovered the break-even ratio $c = 1.5625$ for the illustrative parameters.
- ▶ Kept the exercise distinct from a calibration of the paper.

Extension sketch

Add separate policy wedges to $\kappa_I v_I(n)$ and $\kappa_N v_N(n)$, then study how innovation policy shifts the interior BGP.

The Lean workflow preserves the boundary between source and proof

1. Pin the NBER June 2017 source bytes and inventory the named claims.
2. Translate each selected result into one transparent source-facing Spec.
3. Keep the Lean proof endpoint separate from the semantic statement.
4. Run the paper-scoped build and audit checks.
5. Report partial boundaries instead of turning missing proofs into assumptions.

Paper folder: AR18RaceManMachine

Required check: `python3 scripts/paper_contribution.py check`

`AR18RaceManMachine -fast`

Run status: partially formalized. Proposition 1 and Corollary 1 have source-facing Specs; both proof endpoints contain explicit sorry.

Check status: exit 1. The bounded run reached dependency 173/2505, but no paper module, because the pinned Mathlib checkout had no usable binary cache.

Proposition 1: source claim, Lean Spec, open proof

Original claim under Assumptions 1–3

$$(\exists! E, \text{StaticEq}(D, E)) \wedge \forall E, \text{StaticEq}(D, E) \Rightarrow \text{Eq12}(D, E).$$

Lean statement (ASCII rendering) and exact proof endpoint

```
def proposition1_static_equilibriumSpec (D : StaticModel) : Prop :=
  StaticEnvironment D -> Assumption1 D -> Assumption2 D ->
    Assumption3 D ->
    And (ExistsUnique fun E : StaticOutcome => StaticEquilibrium D E)
      (forall E : StaticOutcome,
        StaticEquilibrium D E -> Equation12Output D E)

theorem proposition1_static_equilibrium (D : StaticModel) :
  proposition1_static_equilibriumSpec D := by
  sorry
```

My reading: ExistsUnique encodes existence and uniqueness; the second conjunct applies equation (12) to every equilibrium. The source-facing statement is explicit, but sorry means Lean has not verified its proof.

Where I did not believe the AI

Claim to test

“Automation necessarily reduces wages and the labor share.”

Verdict

The labor-share statement is correct in the technology-constrained static region. The wage statement is incomplete.

$$\frac{d \ln W}{dI} = P_I - D_I$$

Automation raises the wage when cost savings are large enough. On the interior BGP, capital adjustment makes the long-run wage rise even as employment and the labor share fall.

Source: Own hand derivation of Proposition 3; NBER pp. 12–14

Hand check: sign condition and numerical threshold

When does automation raise wages? - prop. 3

$$I^* = I < \tilde{I}, \quad dN = 0.$$

$$d \ln W = d \ln Y / k_L - (1 - s_L) \frac{\Delta I}{\hat{\sigma} + \varepsilon_L} dI.$$

dividing by dI and subs:

$$\frac{d \ln W}{dI} = \frac{B \hat{\sigma}^{-1}}{1 - \sigma} \left[\underbrace{\left(\frac{W}{f(I^*)} \right)^{1 - \hat{\sigma}} - R^{1 - \hat{\sigma}}}_{\text{productivity effect}} - \underbrace{(1 - s_L) \frac{\Delta I}{\hat{\sigma} + \varepsilon_L}}_{\text{displacement effect}} \right]$$

$$\text{then, } \frac{d \ln W}{dI} > 0 \iff P_I > D_I$$

IF YOU PUT NUMBERS:

$$B = R = 1, \quad \hat{\sigma} = 1.5, \quad s_L = 0.6, \quad \varepsilon_L = 0.5$$

$$\Delta I = 2$$

$$\text{then, } D_I = (1 - 0.6) \frac{2}{1.5 + 0.5} = 0.4$$

$$\text{defining } c = \frac{W / f(I^*)}{R} \quad \text{we get: } P_I = 2(1 - c^{-1/2})$$

ΔI equilibrium:

$$2(1 - c^{-1/2}) = 0.4 \rightarrow c^{-1/2} = 0.8$$

$$\boxed{c = 1.5625}$$

automation does NOT necessarily raise wages. Wages rise when productivity effect is ~~greater~~ greater than D.E

Source: Owner's handwritten derivation of Proposition 3

What the model changes about the automation debate

- ▶ Productivity gains do not determine the wage, employment, or labor share by themselves.
- ▶ Automation displaces labor from existing tasks. New tasks reinstate labor in the task frontier.
- ▶ The short-run wage effect is conditional on productivity gains relative to displacement.
- ▶ An interior innovation race can be stable, but full automation remains possible.
- ▶ Lean is useful here because every domain restriction and inequality must be stated explicitly.

github.com/amchavez/ai-06-restrepo